Name: Vidhan Ghuikhedkar Class: SOC9 Batch: C
Subject: Programming Language Roll No.: 60 Exp. No.: 01
Name of the Experiment: Assignment - 1

Performed on:
Submitted on:

Marks	Teacher's Signature with date
10	

1) List the key features and advantages of Python. Also mention any two python IDE or editors used for program development.

→ i) Simple and Easy to learn – Uses simple English - like syntax.
ii) Interpreted language - Code is executed line by line.
iii) High level language – No need to manage memory manually.
iv) Object - oriented - Supports classes and objects
v) Portable – Same program run on windows, Linux, Mac.
vi) Large library support - Has built-in modules for many task.

Advantages of python → 1) Faster program development
2) fewer lines in code
3) Used in web development, data science, AI, ML
4) free and open - source

Python IDEs and editors
1) IDEs → Default python IDE, simple and beginner friendly.
2) VS code → Popular editor with extension for python.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

2) Explain the history and evolution of python? why is Python popular?

→ History and Evolution of Python →

1) Python was developed by "Guido Van Rossum" in 1991.
2) Named after the tV show "Monty Python's flying circus".
3) It was designed to be simple, readable and powerful.
4) Major version → Python 2x (older), python 3x (current & improved)

Python is popular because →

1) It is easy for beginners
2) It is used in industries
3) Its support many fields: AI, Data science, web development
4) It has large community support

3) Write a Python program to accept two numeric value from the user and perform the following operation.

→
```
a = float (input ("Enter first number:"))
b = float (input ("Enter second number:"))
print ("Addition:", a+b)
print ("Substraction:", a-b)
print ("multiplication:", a*b)
print ("Division:", a/b)
```

Analyze the following python code and identity the data types of each variable. Also explain the output.

1) a = 10
2) b = 5.5
3) c = 3+4j
4) d = True
5) Print (a+b, type(c), d)

→ Data Types

Variable	Values	Data type
a	10	Integer
b	5.5	float
c	3+4j	complex
d	True	Boolean

Output :- 15.5 , <class 'complex'> True
Explanation :-
a+b = 10+5.5 = 15.5
type (c) show complex data type
d print True.

5) Evaluate the use of relational, logical and bitwise oper in Python. Justify with suitable example where each type of operator is most ~~opp~~ appropriate?

→ i) Relational operator :- Used to compared value

a=10	a=10
b=5	b=5
print (a>5)	print (a<b)
output → True	output → False

ii) Logical operator :- Used to combine conditions

x= True	x= True
y = False	y = True
print (x and y)	print (x and y)
output : False	output : True

(iii) Bitwise operators → Used at bit level

```
a = 5   # 0101
b = 3   # 0011
print (a & b)
```

output :- 1

6) Design and implement a menu-driven Python program that:
Accept user input
Perform arithmetic, relational, logical and assignment experation.
Display result clearly.
The program should handle int, float and boolean data type

→
```
a = int (input ("Enter first number:"))
b = int (input ("Enter second number:"))
print ("1 Addition")
print ("2 Greater than check")
print ("3 Logical AND")
print ("4 Assignment operation")
choice = int (input ("Enter your choice"))
if choice == 1:
    ~~print ("a>b:", a>b)~~ print ("Addition:", a+b)
else if choice == 2
    print ("a > b:", a > b)
else if choice == 3
    print ("Logical AND:", a>0 and b>0)
else if choice == 4
    a += b
    print ("After assignment, a =", a)
else:
    print ("Invalid choice")
```

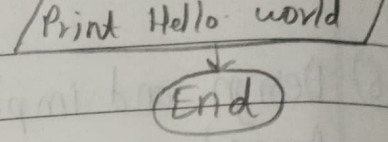
7) Write a program in python to display "Hello World".

→ print ("Hello World!")

output:- Hello World!

Algorithm:- 1) Start
2) Print the message "Hello World"
3) Stop

Flow chart →



8) Write a program code to perform average of 5 number using () function and type conversion concept.

→ Code:-

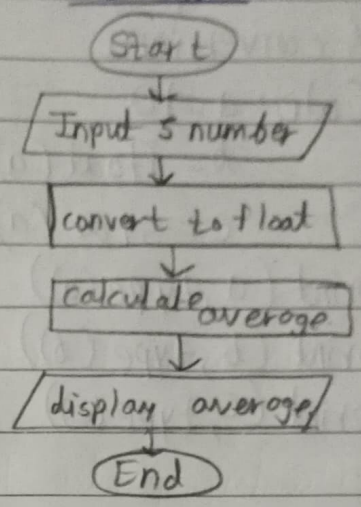
```
a = float (input ("Enter number 1:"))
b = float (input ("Enter number 2:"))
c = float (input ("Enter number 3:"))
d = float (input ("Enter number 4:"))
e = float (input ("Enter number 5:"))
average = (a+b+c+d+e)/5
print ("Average:", average)
```

output:-

Enter number 1 = 1
Enter number 2 = 2
Enter number 3 = 2
Enter number 4 = 3
Enter number 5 = 6
Average = 2.8

* Algorithm:-
1) Start
2) Read five number using input()
3) Convert inputs to float
4) calculate average
5) Display average
6) End

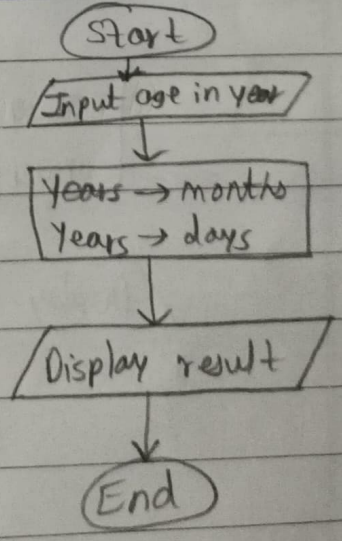
* Flow chart →



9) Write a program that prompts the user to enter their age and print out their age in year, months, and days.

→ age = int(input ("Enter your age in years:"))
maths = age * 12
days = age * 365
print ("Age in years:", age)
print ("Age in months:", months)
print ("Age in days:", days)

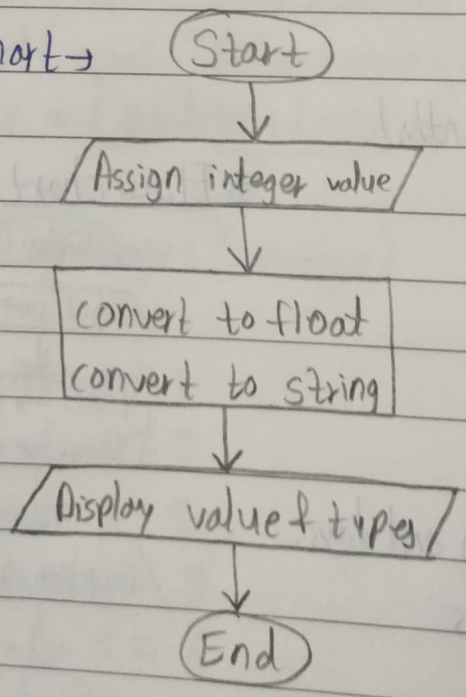
* Flow chart:-



* Algorithm:-
1) Start
2) Read age in years
3) Convert years into months and days.
4) Displays years, months, days
5) stop.

10) Write a program to demonstrate the concept of data type & conversion.

→ Code:
```
a=10
b=float(a)
c=str(a)
print(a,type(a))
print(b,type(b))
print(c,type(c))
```

* Algorithm:- 1) Start
2) Assign integer value
3) Convert integer to float and string
4) Display value and their data type
5) End

* flow chart →